Eenheid 2.7

Note Title 2016/03/30

Die afgeleide as 'n funksie

Stewart p 152

Definisie:

Laat $y = f(x)$ 'n funksie wees.
Die afgeleide van $y = f(x)$ is die funksie

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

met
$D_{f'} = \{x \mid x$ is in die D_f en $f'(x)$ bestaan$\}$

VOORBEELD: Bepaal $f'(x)$ as $f(x) = \sqrt{x}$.

$D_f = [0, \infty)$

- $D_f =$
- $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

Plan??

$$= \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \times \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$$

$$= \lim_{h \to 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \lim_{h \to 0} \frac{\cancel{h}}{\cancel{h}(\sqrt{x+h} + \sqrt{x})} = \cancel{\lim_{h \to 0}} \frac{1}{2\sqrt{x}}$$

Onthou: $D_f = \{x \mid x \geq 0\}$ en $f'(x) = \frac{1}{2\sqrt{x}}$

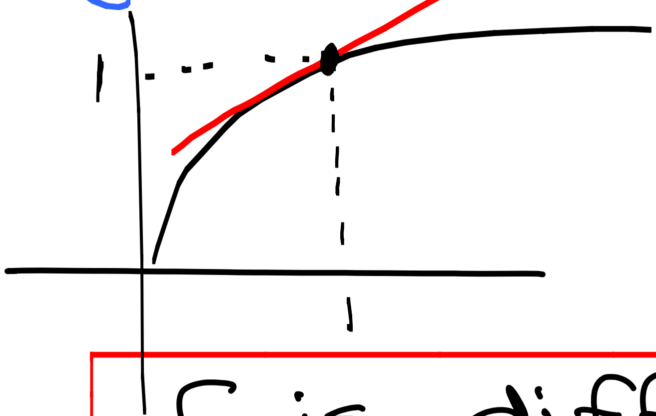
- $D_{f'} = (0, \infty)$

$D_{f'}$ is dus bevat in D_f

Meetkundige interpretasie

$f'(x) = \frac{1}{2\sqrt{x}}$ $\underbrace{f'(1)}_{(m)} = \frac{1}{2\sqrt{1}} = \frac{1}{2}$

- $y = \sqrt{x}$



Helling van raaklyn by $(1, f(1))$ is $\frac{1}{2}$ (eindige helling = getal)

$\boxed{\text{f is differensieerbaar in } x = a \text{ as jy 'n raaklyn met eindige helling kan trek by die punt } (a, f(a))}$

- as die grafiek van $f(x)$ gegee is, kan ons grafiek van f' daarvan aflei en skets.

NB

f' funksie

NB

***3 Vrae beantwoord**

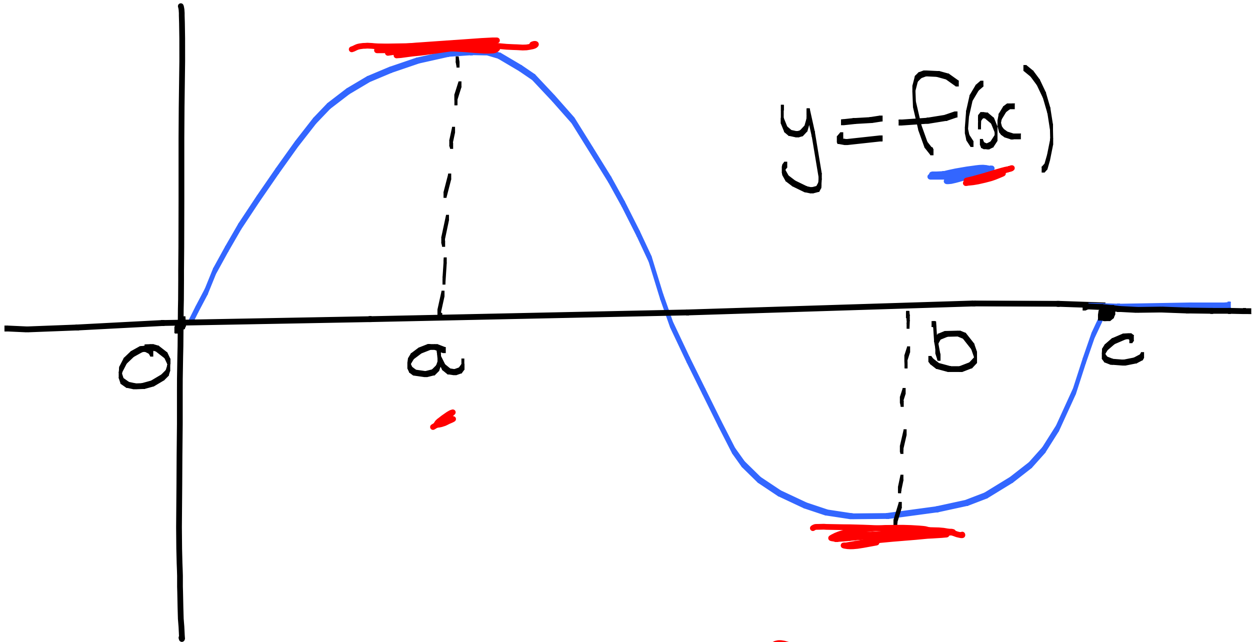
Waar is:

- $f'(x) = 0$? ◁ f 'n horisontale raaklyn
- $f'(x) > 0$? ◁ f styg
- $f'(x) < 0$? ◁ f daal

- Ons stel net belang waar f styg of daal! Op die stadium stel ons nie belang in vorm (f konkaaf op / af) nie

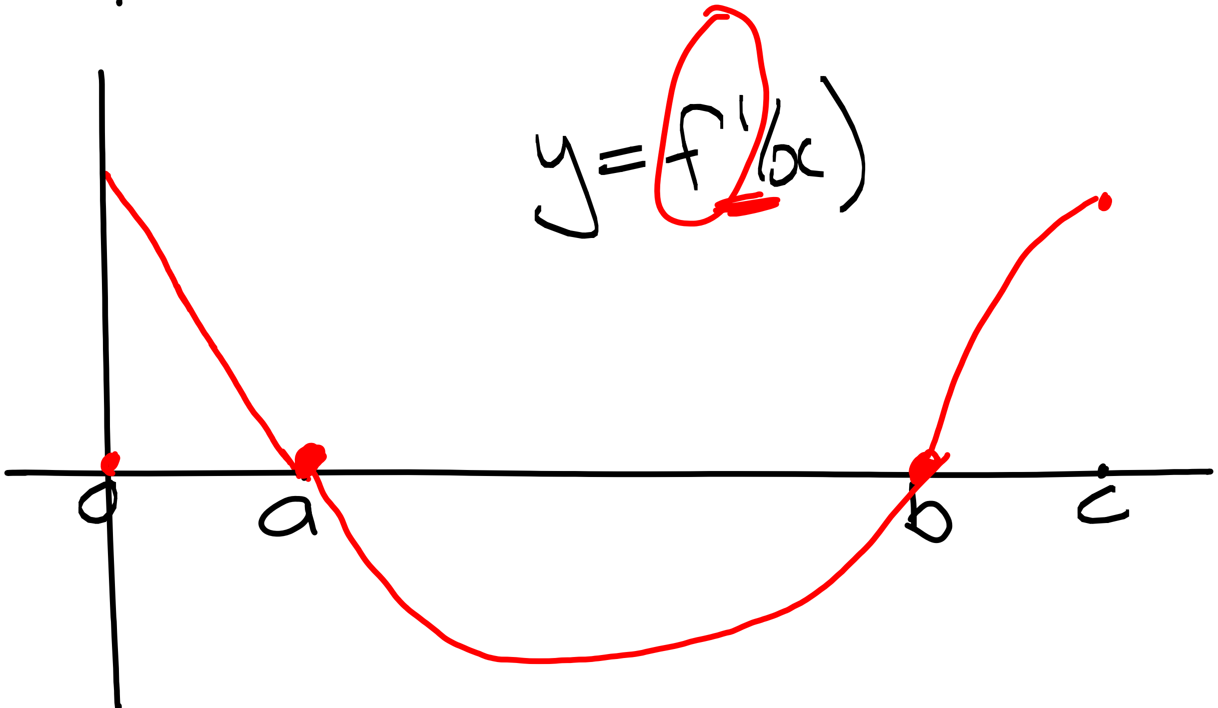
Gegee: $y=f(x)$. Skets vir $f'(x)$

(Dink aan f' as 'n funksie)



• 3 Vrae !!

• $f'(x)=0$? : waar f horisontale raaklyne het by $x=a$ en $x=b$

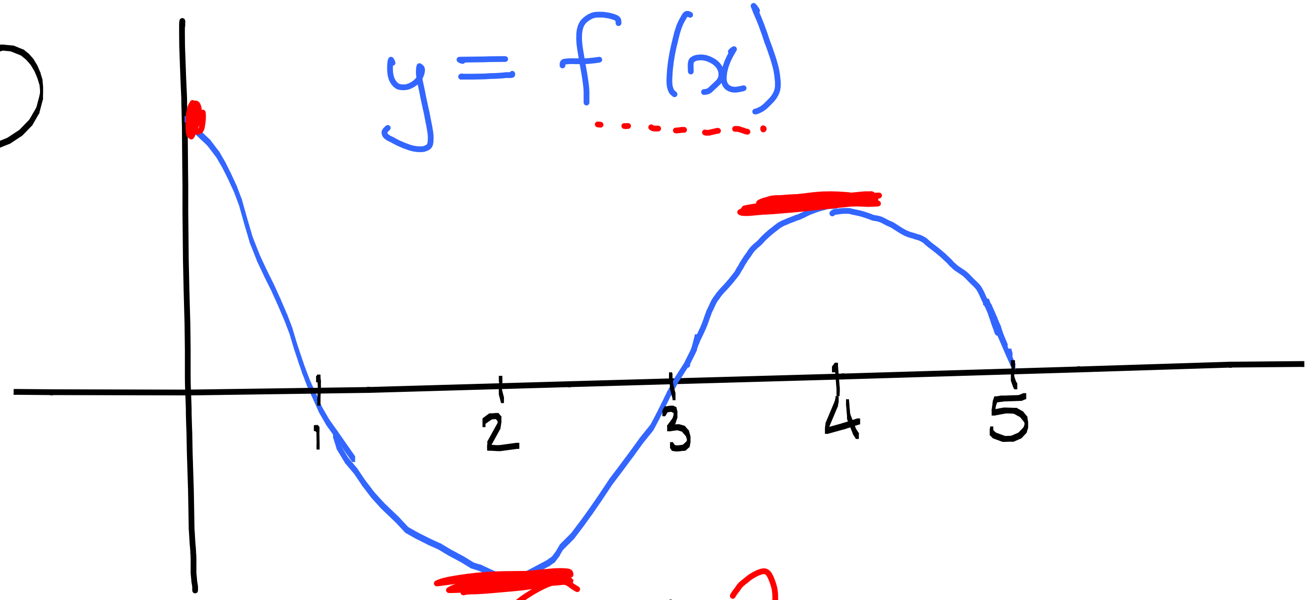


• $f'>0$? : waar f styg
$(0,a)$ en (b,c)

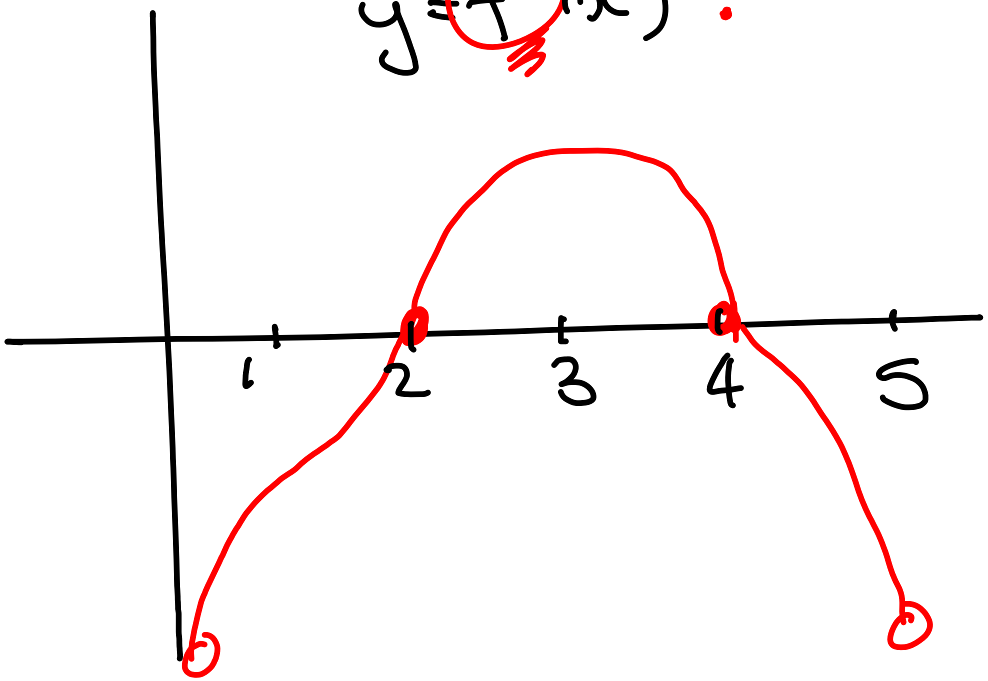
• $f'<0$? : waar f daal
(a,b)

Skets f' as die grafiek van f gegee is

(1)



$y = f'(x)$?

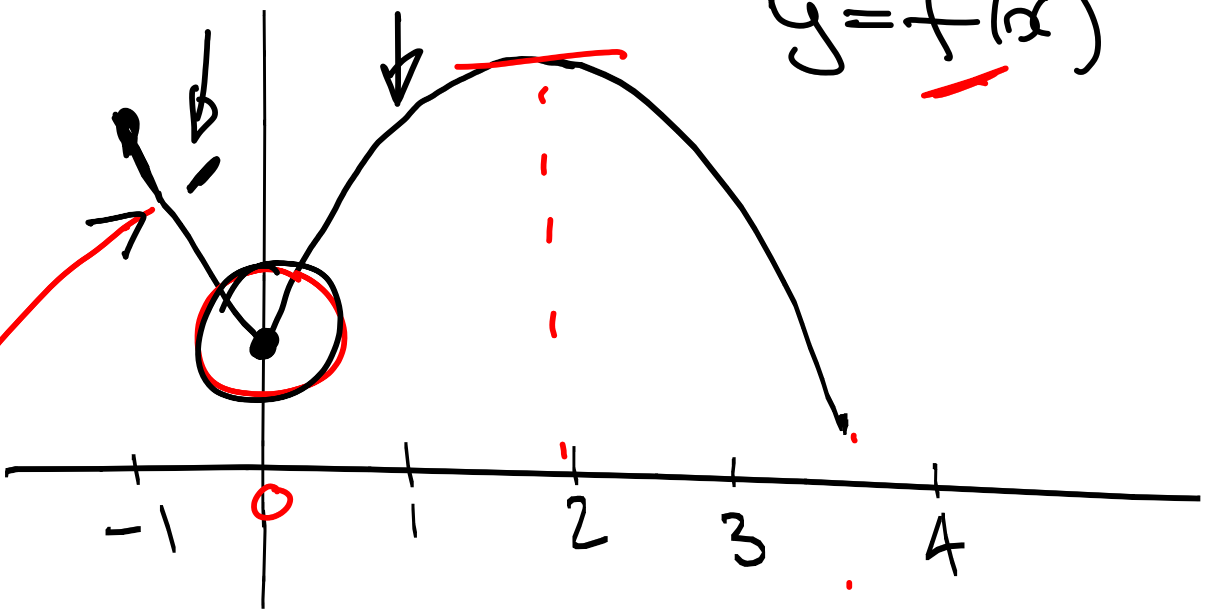


3 Vrae !!

- $f'(x) = 0$ by $x = 2$ en $x = 4$ (f het horisontale raaklyne)
- $f' > 0$ bo x-as waar f styg op $(2,4)$
- $f' < 0$ onder x-as waar f daal op $(0,2)$ en $(4,5)$

2

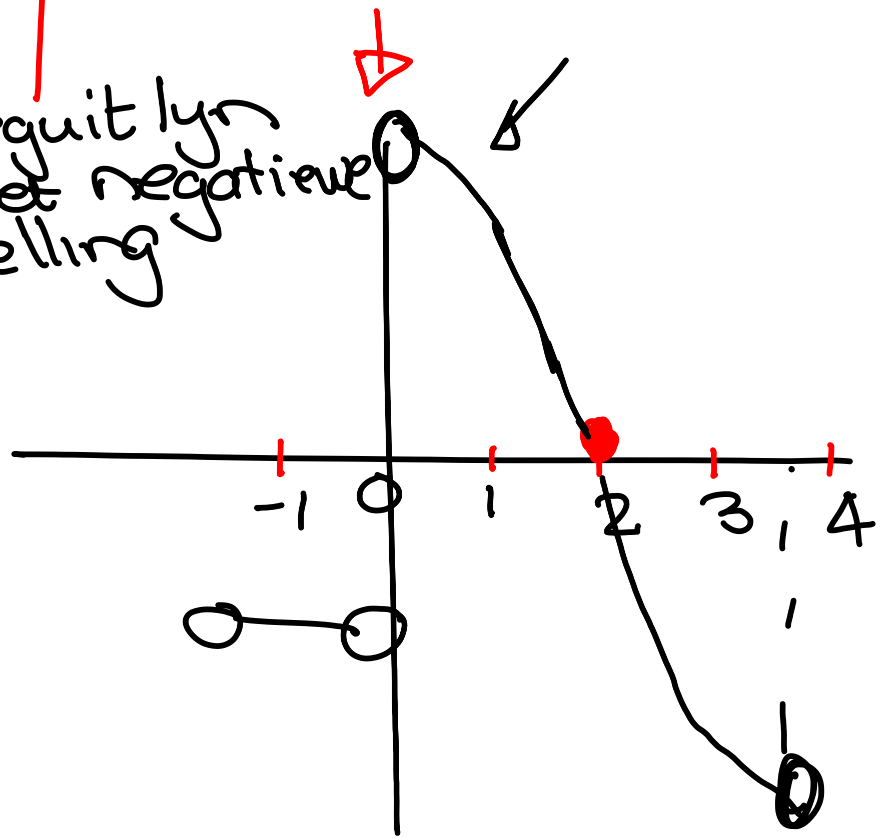
$y = f(x)$



- "hoek" Dan bestaan $f'(0)$ nie NB
- Reguit lyn se afgeleide is 'n konstante

Bv. $f(x) = x$
$f'(x) = 1$

Reguit lyn met negatiewe helling



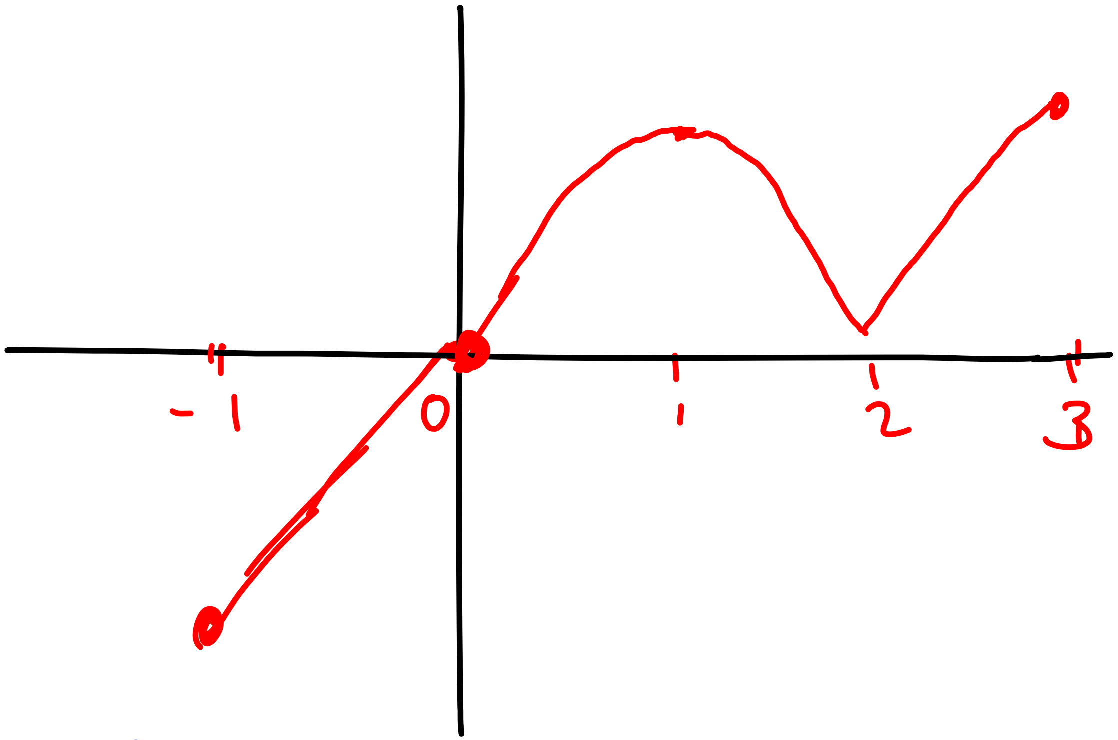
$y = f'(x)$ (3 vrae)

- $f'(x) = 0$ by $x = 2$
(f het horisontale raaklyn)
- $f' > 0$ waar f styg op $(0, 2)$
- $f' < 0$ waar f daal $(-1, 0)$ en $(2, 3.5)$

③ Skets die funksie f wat kontinu is op $[-1,3]$ met:

- $f(0) = 0$
- $f'(0) > 0$ positiewe helling
- $f'(1) = 0$ horisontale raaklyn
- $f'(2)$ bestaan nie

$y = f(x)$



Differensiasie is die proses om 'n afgeleide te bepaal. p 155 (Stewart)

Notasie en Reëls (Leibniz 1684)

$$f'(x) = y' = \frac{dy}{dx}$$

$$f'(x) = \frac{df}{dx} = \frac{d}{dx} f(x)$$

$$f'(x) = Df(x) = D_x f(x)$$

Definisie: 'n Funksie f is differensieer-baar in a as $f'(a) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ bestaan. (dws antwoord is 'n konstant)

Selfstudie p 155

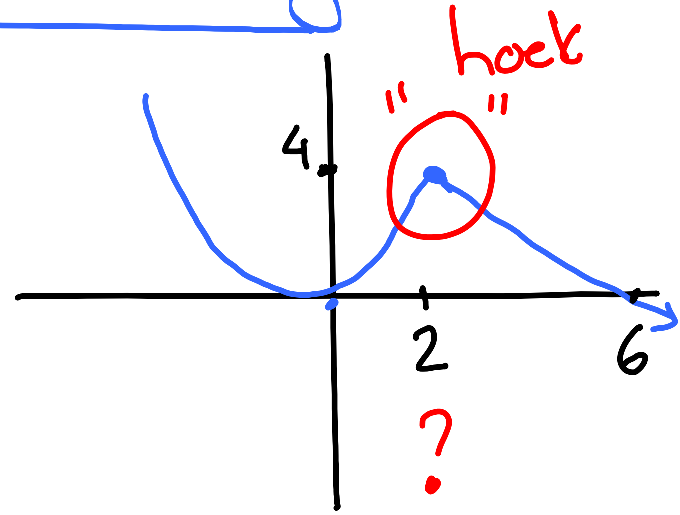
f is differensieerbaar op 'n oop interval (a,b) as dit differensieerbaar is in elke punt in die interval.

NB Wanneer is 'n funksie nie differensieerbaar in 'n punt nie?

3 moontlikhede!!

① **Die funksie het 'n "hoek" by die punt $x=2$.**

$$f(x) = \begin{cases} x^2 & \text{as } x<2 \\ -x+6 & \text{as } x \geq 2 \end{cases}$$



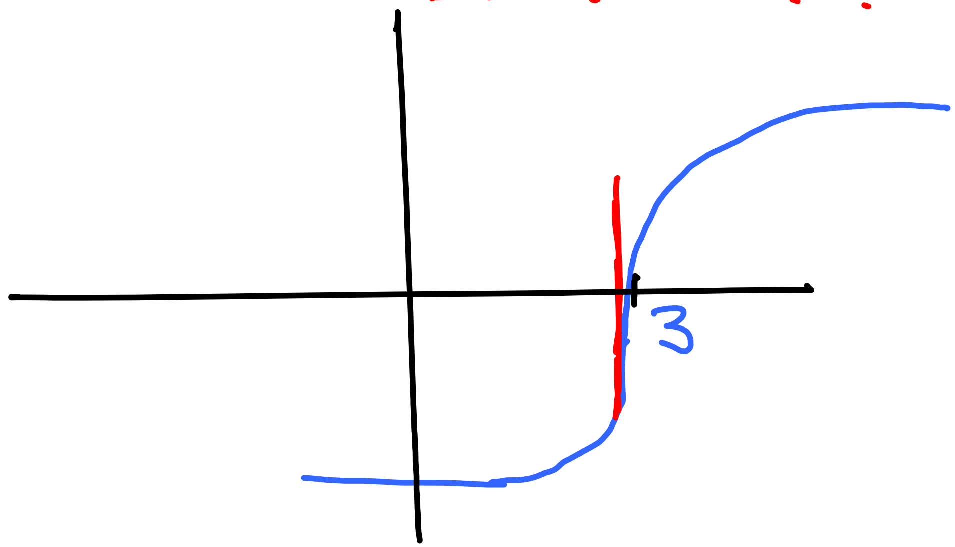
$$f'(x) = \begin{cases} 2x & \text{as } x<2 \\ -1 & \text{as } x>2 \end{cases}$$

$$f'(2) = \begin{cases} 4 \\ -1 \end{cases}$$

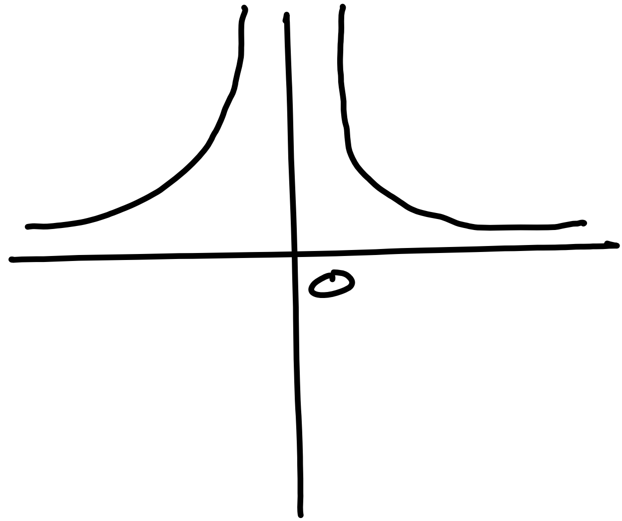
$$f'_L \neq f'_R$$

② $f(x)=(x-3)^{1/3}$ kan 'n vertikale raaklyn by $x=3$ hê.

$f'(3)$ bestaan nie



③ $f(x) = \frac{1}{x^2}$ in $x = 0$



f is diskontinu in $x=0$
want $f(0)$ bestaan nie.

f is nie diffb. in $x=0$

Stelling $\boxed{4}$ p158

As f differensieerbaar is in $x=a$,
dan is f kontinu in $x=a$.

Te Bewys: $*\ \overset{LK}{\lim_{x\to a} f(x)} \overset{?}{=} \overset{RK}{f(a)}$ $*$

(f is kontinu in $x=a$)

Gegee: f is differensieerbaar in $x=a$.

- Dan bestaan $f'(a) = \lim_{x \to a} \frac{f(x)-f(a)}{x-a}$?

Onthou: As dit bestaan is die antwoord 'n konstant.

Bewys: LK

$$= \lim_{x \to a} f(x)$$

$$= \lim_{x \to a} \left[f(x) - f(a) + f(a) \right]$$

$$= \left[\lim_{x \to a} f(x) - f(a) \right] + \cancel{\lim_{x \to a}} f(a) \rightarrow \text{getal}$$

??

$$= \left[\lim_{x \to a} f(x) - f(a) \times \frac{x-a}{x-a}\right] + f(a)$$

$$= \left[\lim_{x \to a} \frac{f(x) - f(a)}{x-a} \times \lim_{x \to a} (x-a)\right] + f(a)$$

gegee

$$= f'(a) \times 0 + f(a)$$

$$= f(a) = Rk$$

konstant

Onthou $\lim_{x \to a} f(x) = f(a)$

$$\therefore \lim_{x \to a} f(x) = f(a)$$

Die ongekeerde van die Stelling is VALS

f kontinu $\Rightarrow$ f differensieerbaar?

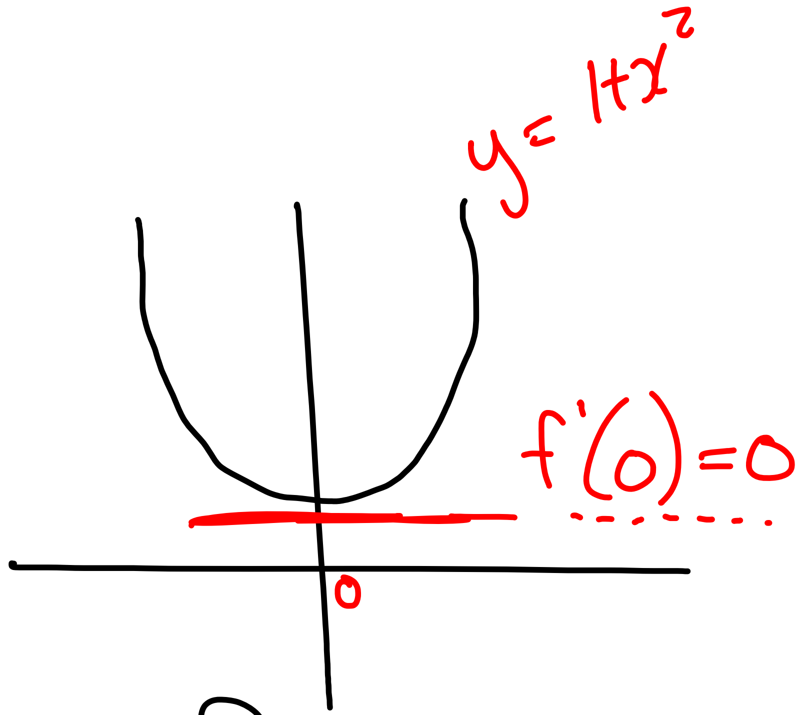
Jy moet toets!!

---

1. $f(x) = 1 + x^2$

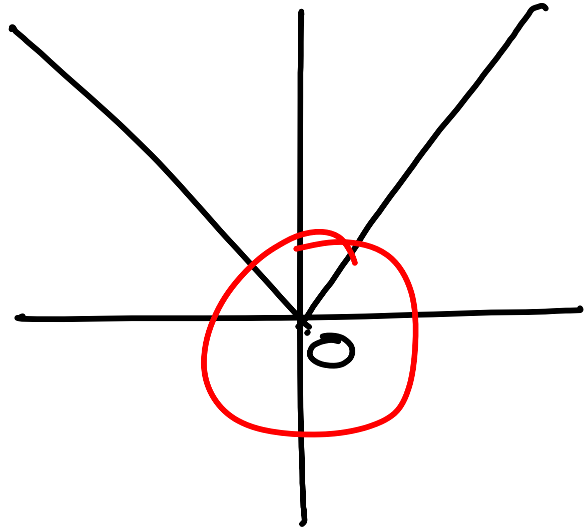
$f'(x) = 2x$

$f'(0) = 0$



f is kontinu in $x=0$ en f is diffb. in $x=0$

② $g(x) = |x|$



g is kontinu in $x=0$

Maar g is nie diffb in $x=0$ nie.

---

bl. 159 **Hoër Afgeleides**

$$f''(x) = \frac{d}{dx}\left(f'(x)\right) = \frac{d^2}{dx^2}\left(f(x)\right)$$

$$f^n(x) = \frac{d^n}{dx}\left(f(x)\right)$$

- Uit Prakties: Ex 2.3 # 59 bl. 102

As $\lim_{x \to 1} \frac{f(x) - 8}{x - 1} = 10$ bewys dat $\lim_{x \to 1} f(x) = 8$

gegee gebruik

LK RK

Bewys: LK: $\lim_{x \to 1} \left[f(x) - 8 + 8 \right]$

$= \lim_{x \to 1} \left[f(x) - 8 \right] + \cancel{\lim_{x \to 1}} 8$

$= \lim_{x \to 1} \left[f(x) - 8 \times \frac{x-1}{x-1} \right] + 8$

$$= \lim_{x\to 1}\left[\frac{f(x)-8}{x-1}\right] \times \lim_{x\to 1}(x-1) + 8$$

$\checkmark$ gegee

$$= 10 \times \lim_{x\to 1}(x-1) + 8$$

$$= 10 \times 0 + 8$$

$$= 8 = Rk$$